



Chapter 1: Linear equations in Linear Algebra

Slide 18:

$$\begin{bmatrix} 0 & 1 & 1 & 4 \\ 3 & 6 & -3 & 3 \\ -2 & -3 & 7 & 10 \end{bmatrix} \sim \begin{bmatrix} -2 & -3 & 7 & 10 \\ 3 & 6 & -3 & 3 \\ 0 & 1 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} -2 & -3 & 7 & 10 \\ 1 & 2 & -1 & 1 \\ 0 & 1 & 1 & 4 \end{bmatrix}$$

Interchange R1 and R3 R2 = R2 * (1/3)

$$\begin{bmatrix} -2 & -3 & 7 & 10 \\ 0 & 1 & 5 & 12 \\ 0 & 1 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} -2 & -3 & 7 & 10 \\ 0 & 1 & 5 & 12 \\ 0 & 0 & -4 & -8 \end{bmatrix} \text{ REF Consistent with unique solution}$$

R2=R2 + (1/2)R1 R3=R3-R2

$$\begin{bmatrix} -2 & -3 & 7 & 10 \\ 0 & 1 & 5 & 12 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} -2 & -3 & 0 & -4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} -2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

R3=R3/-4 R1=R1-7R3 & R2= R2-5R3 R1=R1+3R2

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \text{ RREF } x=-1, y=2, z=2$$

R1=R1/-2

Slide 18: (second system)

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 2 & 6 & -4 & 2 \\ 3 & 9 & -6 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 3 & -2 & 1 \\ 1 & 3 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

R2=R2-2R1 & R3=R3-3R1 infinite solutions

X2 and x3 are free

$$x = \begin{cases} -3x_2 + 2x_3 + 1 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases} \text{ parametric representation}$$

Slide 18: (third system)

$$\begin{aligned} & \begin{bmatrix} 0 & 1 & 1 & 4 \\ 3 & 6 & -3 & 3 \\ -2 & -3 & 3 & 2 \end{bmatrix} \xrightarrow{\text{Interchange R1 and R3}} \begin{bmatrix} -2 & -3 & 3 & 2 \\ 1 & 2 & -1 & 1 \\ 0 & 1 & 1 & 4 \end{bmatrix} \xrightarrow{\text{R2=R2+(1/2)R1}} \begin{bmatrix} -2 & -3 & 3 & 2 \\ 0 & \frac{1}{2} & \frac{1}{2} & 2 \\ 0 & 1 & 1 & 4 \end{bmatrix} \\ & \xrightarrow{\text{R2=R2*2}} \begin{bmatrix} -2 & -3 & 3 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 1 & 1 & 4 \end{bmatrix} \xrightarrow{\text{R3=R3-R2 \& R1=R1+3R2}} \begin{bmatrix} -2 & 0 & 6 & 14 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{R1=R1/-2}} \begin{bmatrix} 1 & 0 & -3 & -7 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ RREF} \end{aligned}$$

Consistent with infinite number of solutions

$$X = \begin{cases} -7 + 3x_3 \\ x_2 = 4 - x_3 \\ x_3 \text{ is free} \end{cases}$$

Slide 18: (fourth system)

Surely consistent as the system is homogenous

$$\begin{aligned} & \begin{bmatrix} 1 & 3 & 0 & -1 & 0 \\ -3 & -9 & 1 & 1 & 0 \\ 2 & 6 & 1 & -4 & 0 \end{bmatrix} \xrightarrow{\text{R2=R2+3R1 \& R3=R3-2R1}} \begin{bmatrix} 1 & 3 & 0 & -1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & -2 & 0 \end{bmatrix} \xrightarrow{\text{R3=R3-R2}} \begin{bmatrix} 1 & 3 & 0 & -1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

It has an infinite number of solutions as x_2 and x_4 are free

$$\begin{cases} x_1 = -3x_2 + x_4 \\ x_2 \text{ is free} \\ x_3 = 2x_4 \\ x_4 \text{ is free} \end{cases}$$

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$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 5 \\ 3 & 6 & -3 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

last row shows that the system is inconsistent